

Venus

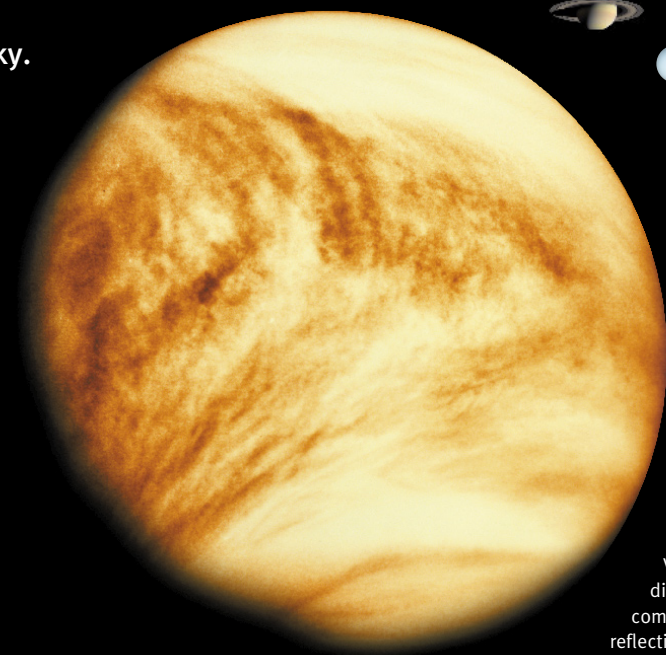
Venus is the second rocky planet from the Sun and the brightest in Earth's sky. It is Earth's inner neighbor and twin, with a similar size and mass. However, Venus has lost all of its water, and its carbon dioxide has escaped to form a very dense atmosphere that acts like a greenhouse, holding in heat.

FEATURES

The similarity between Venus and Earth means that, like Earth, Venus has a hot rocky mantle beneath its solid surface crust. This is the source of Venus's spasmodic volcanic activity. Below this is an iron-nickel core, which has a central solid region surrounded by a liquid-metal outer region. Due to the baking-hot temperature at Venus's surface, the vast majority of the water that was released from the mantle rocks in the past has escaped into space. As a result, the dry mantle rocks are very viscous, and Venus has no moving plates and no mountain ranges.

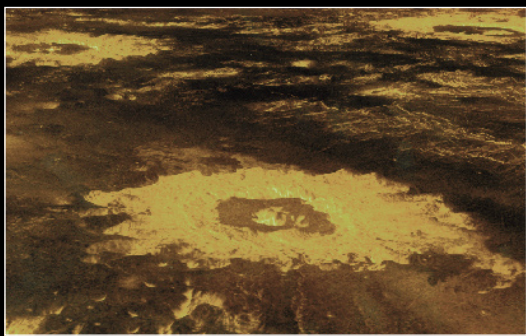
Radar systems onboard orbiting spacecraft have penetrated Venus's clouds and mapped its surface. It is dominated by volcanic features: 85 percent of it is covered with low-lying plains of lava produced about 500 million years ago, and hundreds of volcanoes are visible. Large impact craters also dot the surface.

Venus is the slowest-spinning planet, taking longer to spin than to complete an orbit of the Sun. Uniquely, it spins from east to west.



Cloudy world

Venus has a dense carbon-dioxide atmosphere and is completely shrouded by thick, highly reflective clouds made of weak sulfuric acid droplets. These start about 28 miles (45 km) above the surface and extend up to a height of 43 miles (70 km). Below the clouds is an overcast, orange-colored world.



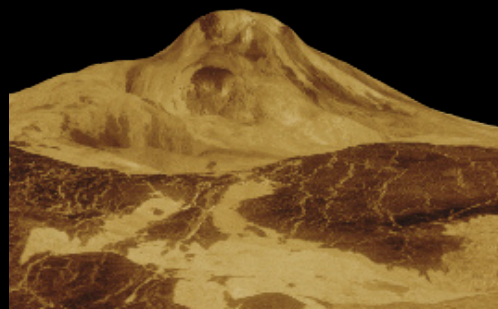
Saskia Crater

Impact craters on Venus vary from 168 miles (270 km) down to 4 miles (7 km) across. Saskia is middle-sized, at 23 miles (37 km), and has a central mountain. This radar view has been colored based on other images taken on the surface.



Wind streaks

The surface of Venus is windy, and since the winds generally blow in only one direction, wind streaks form. The streaks of dust and soil shown here are 22 miles (35 km) long.



Maat Mons

This large shield volcano rises about 3 miles (5 km) above the surrounding plains. Lava has poured out of the volcano, spread out in all directions for hundreds of miles, and then solidified.

PLANET DATA

DIAMETER 7,521 miles (12,104 km)

AVERAGE DISTANCE FROM SUN
67.2 million miles (108.2 million km)

ORBITAL PERIOD 224.7 Earth days

ROTATION PERIOD 243 Earth days

SURFACE TEMPERATURE
867°F (464°C)

NUMBER OF MOONS none



Binocular view

Venus is extremely bright when viewed through binoculars—its crescent phase is apparent—but all that is seen are the tops of the thick, cloudy blanket.



Phases of Venus

Because its distance from Earth changes with its various phases, Venus appears three times bigger when in crescent phase than it does when full.

Transit of Mercury

Mercury's orbit usually takes it above or below the Sun's disk in the sky. Infrequently, it is seen to transit the Sun's face. It can take up to 9 hours for Mercury to cross the face of the Sun. The black planetary dots on this image are Mercury's transit on May 7, 2003. The transit shows just how small Mercury is in comparison to the Sun. The next transit will take place on May 9, 2016.

